

Lab Assignment 3

Servlet lifecycle with HttpServlet/GenericServlet implementation

1. Title

Implementation of Servlet Lifecycle and Scrollable ResultSet Navigation using JDBC

2. Aim

To study the **Servlet lifecycle using GenericServlet and HttpServlet** and implement **Scrollable ResultSet navigation and update operations using JDBC** in console and Swing applications.

3. Learning Objectives

- Understand Servlet lifecycle methods
- Differentiate between **GenericServlet** and **HttpServlet**
- Create **Scrollable ResultSet**
- Navigate records using cursor methods
- Update records using `updateRow()`
- Develop Swing GUI for record navigation
- Perform database updates through GUI

4. Software Requirements

- JDK 8 or above
- Eclipse / NetBeans / IntelliJ
- MySQL Server
- MySQL Connector/J Driver
- Apache Tomcat Server
- Database: **college**
- Table: **student**

5. Database Setup

Step 1: Create Database

```
CREATE DATABASE exp;  
USE college;
```

Step 2: Create Table

```
CREATE TABLE student (  
    id INT PRIMARY KEY,  
    name VARCHAR(50),  
    course VARCHAR(30),  
    marks INT  
);
```

3.1. Write a console program that:

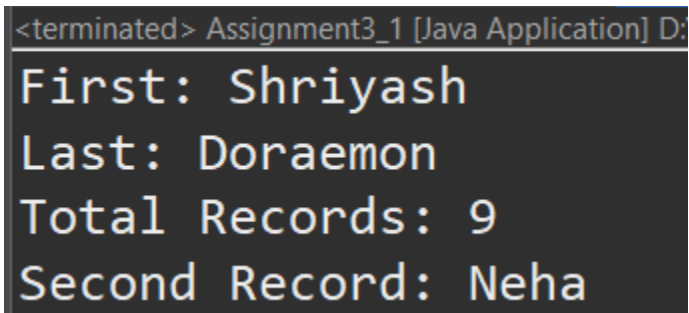
- Creates a Scrollable ResultSet

- Displays first record
- Displays last record
- Moves to 2nd record using `absolute(2)`
- Displays total number of records

Code:

```
import java.sql.*;
public class Assignment3_1 {
    public static void main(String[] args) {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/exp", "root",
"root");
            Statement stmt = con.createStatement(ResultSet.TYPE_SCROLL_INSENSITIVE,
ResultSet.CONCUR_READ_ONLY);
            ResultSet rs = stmt.executeQuery("SELECT * FROM student");
            if (rs.first()) System.out.println("First: " + rs.getString("name"));
            if (rs.last()) {
                System.out.println("Last: " + rs.getString("name"));
                System.out.println("Total Records: " + rs.getRow());
            }
            if (rs.absolute(2)) {
                System.out.println("Second Record: " + rs.getString("name"));
            }
            con.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

Output:



```
<terminated> Assignment3_1 [Java Application] D:
First: Shriyash
Last: Doraemon
Total Records: 9
Second Record: Neha
```

3.2. Write a console program that:

- Moves to a particular record
- Updates marks using `updateInt()`
- Calls `updateRow()`
- Displays updated record

Code:

```
import java.sql.*;

public class Assignment3_2 {
    public static void main(String[] args) {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/exp", "root",
"root");

            Statement stmt = con.createStatement(ResultSet.TYPE_SCROLL_SENSITIVE,
ResultSet.CONCUR_UPDATABLE);
            ResultSet rs = stmt.executeQuery("SELECT * FROM student");

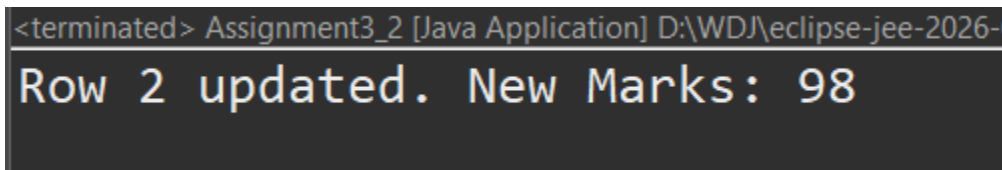
            if (rs.absolute(2)) {
                rs.updateInt("marks", 98);

                rs.updateRow();

                System.out.println("Row 2 updated. New Marks: " + rs.getInt("marks"));
            }

            con.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

Output:

A screenshot of a Java application window. The title bar reads "<terminated> Assignment3_2 [Java Application] D:\WD\eclipse-jee-2026-". The main content area displays the text "Row 2 updated. New Marks: 98" in a large, bold, white font on a dark background.

3.3. Create a Swing application with:

- TextFields: id, name, course, marks
- Buttons:
 - First
 - Previous
 - Next
 - Last

Code:

```
import javax.swing.*.*;
```

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```
import java.awt.*;
import java.sql.*;

public class Assignment3_3 extends JFrame {
    JTextField txtId, txtName, txtCourse, txtMarks;
    JButton btnFirst, btnPrev, btnNext, btnLast;
    ResultSet rs;

    public Assignment3_3() {
        setTitle("Student Navigation");
        setLayout(new GridLayout(5, 2, 10, 10));

        add(new JLabel("ID:")); txtId = new JTextField(); add(txtId);
        add(new JLabel("Name:")); txtName = new JTextField(); add(txtName);
        add(new JLabel("Course:")); txtCourse = new JTextField(); add(txtCourse);
        add(new JLabel("Marks:")); txtMarks = new JTextField(); add(txtMarks);

        btnFirst = new JButton("First"); btnPrev = new JButton("Previous");
        btnNext = new JButton("Next"); btnLast = new JButton("Last");
        add(btnFirst); add(btnPrev); add(btnNext); add(btnLast);

        initDB();

        btnFirst.addActionListener(e -> { try { rs.first(); showData(); } catch(Exception ex){} });
        btnLast.addActionListener(e -> { try { rs.last(); showData(); } catch(Exception ex){} });
        btnNext.addActionListener(e -> { try { if(rs.next()) showData(); else rs.last(); } catch(Exception ex)
        {} });
        btnPrev.addActionListener(e -> { try { if(rs.previous()) showData(); else rs.first(); } catch(Exception
        ex){} });

        setSize(400, 300);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setVisible(true);
    }

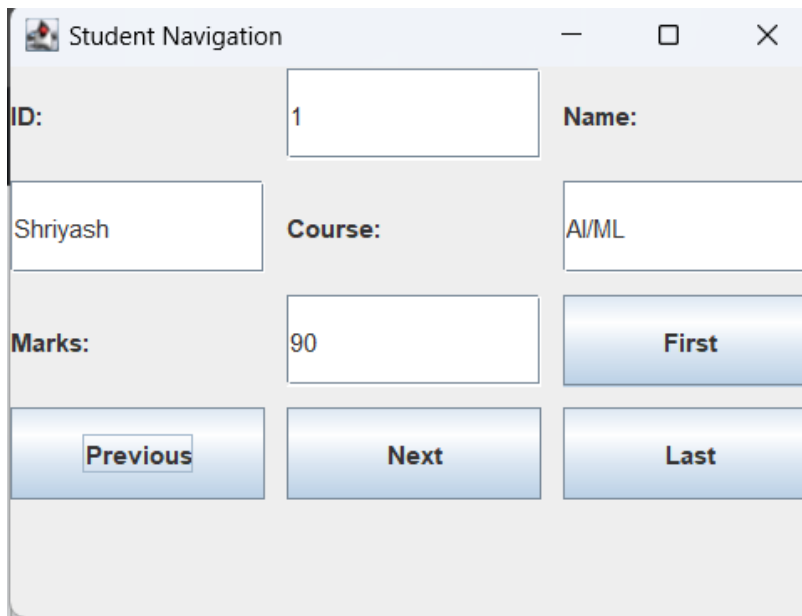
    private void initDB() {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/exp", "root",
            "root");
            Statement stmt = con.createStatement(ResultSet.TYPE_SCROLL_INSENSITIVE,
            ResultSet.CONCUR_READ_ONLY);
            rs = stmt.executeQuery("SELECT * FROM student");
            rs.next(); showData();
        } catch (Exception e) { e.printStackTrace(); }
    }

    private void showData() throws SQLException {
        txtId.setText(rs.getString("id"));
        txtName.setText(rs.getString("name"));
        txtCourse.setText(rs.getString("course"));
    }
}
```

```
        txtMarks.setText(rs.getString("marks"));
    }

    public static void main(String[] args) { new Assignment3_3(); }
}
```

Output:



The screenshot shows a Java Swing window titled "Student Navigation". It features a grid layout with the following elements:

- ID:** A text field containing the value "1".
- Name:** A text field containing the value "Shriyash".
- Course:** A text field containing the value "AI/ML".
- Marks:** A text field containing the value "90".
- Buttons:** Four buttons are present: "First" (top right), "Previous" (bottom left), "Next" (bottom center), and "Last" (bottom right).

3.4. Enhance previous GUI by adding:

- Update button

Code:

```
import javax.swing.*.*;
import java.awt.*.*;
import java.sql.*.*;

public class Assignment3_4 extends JFrame {
    JTextField txtId, txtName, txtCourse, txtMarks;
    JButton btnFirst, btnPrev, btnNext, btnLast, btnUpdate;
    ResultSet rs;

    public Assignment3_4() {
        setTitle("Student Manager PRO");
        setLayout(new GridLayout(6, 2, 10, 10));

        add(new JLabel("ID:")); txtId = new JTextField(); add(txtId);
        add(new JLabel("Name:")); txtName = new JTextField(); add(txtName);
    }
}
```

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```
add(new JLabel("Course:")); txtCourse = new JTextField(); add(txtCourse);
add(new JLabel("Marks:")); txtMarks = new JTextField(); add(txtMarks);

btnFirst = new JButton("First"); btnPrev = new JButton("Prev");
btnNext = new JButton("Next"); btnLast = new JButton("Last");
btnUpdate = new JButton("Update Record");

add(btnFirst); add(btnPrev); add(btnNext); add(btnLast);
add(new JLabel("Action:")); add(btnUpdate);

initDB();

btnFirst.addActionListener(e -> { try { rs.first(); showData(); } catch(Exception ex){} });
btnLast.addActionListener(e -> { try { rs.last(); showData(); } catch(Exception ex){} });
btnNext.addActionListener(e -> { try { if(rs.next()) showData(); } catch(Exception ex){} });
btnPrev.addActionListener(e -> { try { if(rs.previous()) showData(); } catch(Exception ex){} });

btnUpdate.addActionListener(e -> {
    try {
        rs.updateInt("marks", Integer.parseInt(txtMarks.getText()));
        rs.updateRow();
        JOptionPane.showMessageDialog(this, "Data Updated Successfully!");
    } catch (Exception ex) { JOptionPane.showMessageDialog(this, "Update Failed!"); }
});

setSize(400, 350);
setDefaultCloseOperation(EXIT_ON_CLOSE);
setVisible(true);
}

private void initDB() {
    try {
        Class.forName("com.mysql.cj.jdbc.Driver");
        Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/exp", "root",
"root");
        Statement stmt = con.createStatement(ResultSet.TYPE_SCROLL_SENSITIVE,
ResultSet.CONCUR_UPDATABLE);
        rs = stmt.executeQuery("SELECT * FROM student");
        rs.next(); showData();
    } catch (Exception e) { e.printStackTrace(); }
}

private void showData() throws SQLException {
    txtId.setText(rs.getString("id"));
    txtName.setText(rs.getString("name"));
    txtCourse.setText(rs.getString("course"));
    txtMarks.setText(rs.getString("marks"));
}

public static void main(String[] args) { new Assignment3_4(); }
}
```

Output:

The screenshot shows a Java Swing window titled "Student Manager PRO". The window contains a form with the following elements:

- ID:** A text field containing the value "1".
- Name:** A text field containing the value "Shriyash".
- Course:** A text field containing the value "AI/ML".
- Marks:** A text field containing the value "95".
- Action:** A button labeled "Update Record".
- Navigation:** Four buttons labeled "Prev", "Next", "Last", and "First" are arranged in a row.

Conclusion

Thus, the implementation of **Servlet lifecycle using GenericServlet and HttpServlet** was successfully studied and demonstrated. The lifecycle methods such as **init()**, **service()**, **doGet()/doPost()**, and **destroy()** were understood in detail. Additionally, **Scrollable ResultSet navigation** was implemented using JDBC to move across records with cursor methods like **first()**, **last()**, **previous()**, **next()**, and **absolute()**, and records were updated using **updateRow()**. A **Swing-based GUI application** was also developed to navigate and update student records interactively. This practical improved understanding of **Servlet architecture, database connectivity, and dynamic record manipulation using JDBC and Java Swing**.